Can Machine Learning Variable Selection and Individual Commodity Fundamentals Improve Futures Return Forecasting?

Leo Specks
Università Bocconi
Supervisor: Prof. Massimo Guidolin

July 2027

DRAFT
Last Modified: May 17, 2026
Download latest version *here*.

Abstract

Contents

1 Introduction

1.1 Motivation & Background

(Guidolin and Pedio, 2021)

The financialization of commodity markets; why forecasting commodity futures matters to investors, hedgers, and risk managers; the tension between statistical and economic loss functions.

1.2 The Research Gap

Three specific limitations of Guidolin & Pedio (2021): fragility of stepwise AIC selection, use of only aggregate commodity factors rather than individual fundamentals, and a sample that ends in 2012 — missing the COVID shock and post-GFC cycles.

1.3 Research Questions & Hypotheses

This thesis aims at answering whether replacing AIC-based stepwise regression with penalized ML methods (LASSO/Elastic Net) improve the out-of-sample forecasting accuracy of commodity futures returns, both statistically and in portfolio returns. Furthermore, it assesses if commodity-specific individual fundamental predictors (inventory levels, crop reports, weather data) add forecasting power beyond aggregate cross sectional factors on an extended 2004-2024 sample.

1.4 Contributions to the Literature

Methodological contribution (ML selection), empirical contribution (individual fundamentals), and out-of-sample robustness contribution (extended sample).

1.5 Thesis Outline

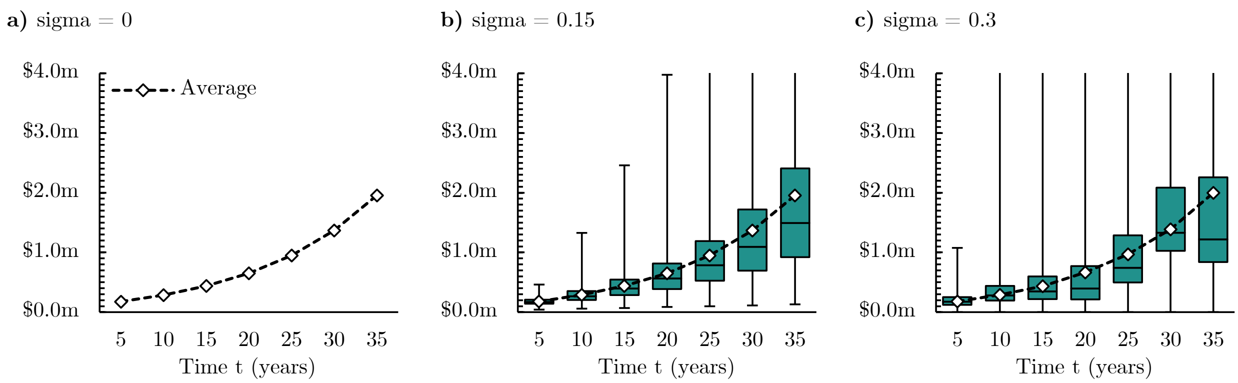
Explain the strucutre of the thesis.

2 Literature Review

2.1 Theories of Commodity Futures Returns

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris consequat sapien sit amet tortor tincidunt, tempus interdum arcu pellentesque. Nunc tellus justo, laoreet eu velit eu, vestibulum varius lorem. Vestibulum laoreet facilisis pretium. In feugiat porta lorem et finibus. Nulla vitae diam lacus. Aliquam vitae molestie felis. Aliquam ac lacus et ante ornare cursus. Duis non eros sem. Donec pellentesque ex metus, ac accumsan metus vulputate vitae. Aliquam erat volutpat. Morbi imperdiet facilisis mollis. Vestibulum consectetur est in justo posuere pretium. Nam lacus lacus, hendrerit sed justo sed, tincidunt fermentum lacus. Suspendisse ultrices libero sit amet lectus eleifend, tincidunt scelerisque massa mattis. Proin sed velit porttitor, condimentum nisi pulvinar, malesuada ex. Nam vehicula tempor augue vel consectetur.

Figure 1
Linear Return $r_{t,t+n}$ of the following indices i) MSCI World, ii) MSCI World Equal Weighted, iii) MSCI World Small Cap, iv) MSCI Europe and v) MSCI Emerging Markets IMI



Nunc volutpat, massa eu dignissim dignissim, purus nunc vestibulum libero, quis bibendum nulla diam sed risus. Mauris elementum turpis nec massa mattis dictum. Nulla ac dolor eu neque consequat suscipit. Fusce sed lorem tellus. Integer efficitur orci sed ligula vestibulum, at sodales odio eleifend. Suspendisse luctus diam at nisi pharetra sollicitudin. Proin est sapien, efficitur vitae sodales sodales, pellentesque in tortor. Duis at commodo diam. Morbi lobortis dictum justo vitae suscipit. Vivamus dictum lacus velit, vitae sagittis lorem sodales eu. Nam a arcu rutrum, aliquam mauris quis, tempus ante. Sed vehicula tortor id arcu dapibus, quis blandit dui egestas. Mauris in ullamcorper augue. Aenean vehicula neque quis felis lacinia, ut pharetra felis vestibulum. Maecenas sit amet lacus scelerisque orci condimentum pellentesque et vehicula lacus. Duis vitae ex a risus molestie pretium. In hac habitasse platea dictumst. Phasellus fringilla sit amet erat quis euismod. Vivamus turpis eros, vulputate id condimentum sit amet, tempus a velit. Interdum et malesuada fames ac ante ipsum primis in faucibus. Donec et convallis tortor. Aenean porttitor fermentum urna ut condimentum.

Integer luctus, ante eu finibus facilisis, turpis dui scelerisque sapien, quis elementum dui tortor vitae nisl. Aliquam mattis mauris in dui semper lacinia. Cras eros tellus, condimentum sed hendrerit quis, vehicula quis risus. In commodo in orci at sodales. Etiam gravida varius ex quis semper. Morbi molestie ipsum id purus faucibus, laoreet cursus tortor sollicitudin. In sodales blandit nunc, vitae ultricies orci bibendum non. Sed dapibus ullamcorper magna, vitae tempor dolor condimentum id. In posuere ex id vehicula tempus. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Aenean auctor justo eget tincidunt condimentum. Integer euismod massa ligula, at imperdiet nisi lobortis in. Nunc porttitor vehicula diam iaculis mattis. Interdum et malesuada fames ac ante ipsum primis in faucibus. Pellentesque posuere ligula id felis vehicula tincidunt. Mauris et mi ex. Morbi nec ligula mollis, consectetur felis id, auctor leo. Vivamus a molestie ante. Suspendisse placerat dolor ex. Nam interdum luctus felis eu eleifend. Mauris ac vulputate ex, id feugiat ipsum. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut ex leo, ullamcorper ut magna eget, tempus consectetur orci. Nunc vitae ante lectus. Ut malesuada diam ut arcu condimentum, at maximus justo ullamcorper. Mauris vehicula tortor in dolor feugiat elementum. Pellentesque tellus ligula, condimentum et felis et, ultrices laoreet turpis. Aliquam quis dolor luctus, iaculis diam vel, sodales dui.

Mauris ultricies arcu ut neque semper, et pretium arcu sagittis. Mauris vestibulum maximus arcu. Sed vulputate convallis bibendum. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Suspendisse posuere arcu sed ultricies gravida. In hac habitasse platea dictumst. Aliquam id erat id nunc luctus bibendum. Curabitur tincidunt leo nec urna posuere vestibulum. Morbi lacinia aliquam arcu, quis interdum tortor congue id. Aliquam semper varius imperdiet. Donec tincidunt magna et dolor volutpat aliquet. Etiam ac pretium ante. Nam quis lacus sit amet metus egestas laoreet. Sed in orci venenatis, laoreet mi non, ullamcorper quam. Nullam eu lobortis nibh. Duis id neque vitae dolor blandit maximus. Vivamus ligula velit, posuere ac tortor non, euismod aliquam nisi. Etiam a ultrices nisi, vitae tincidunt leo. Donec scelerisque felis magna, sit amet euismod ex sodales a. Nunc consectetur purus nunc, vitae volutpat lectus fringilla vitae. Suspendisse ut pharetra justo, non ultrices est. Fusce sit amet ante non sapien mollis elementum id eu augue. Donec consectetur risus eget tellus pellentesque, at maximus mi elementum. Morbi id dapibus ligula. Praesent vel tempor ipsum. Phasellus at justo venenatis, lobortis leo vitae, molestie metus. Cras ullamcorper ante eu mi laoreet, quis varius tellus consectetur.

Table 1

UMD Factor: Summary Statistics (Jan 1927–Feb 2026)

Statistic	Monthly	Annualized
Mean Return (%)	0.612	7.34
Standard Deviation (%)	4.68	16.21
Sharpe Ratio	-	0.453
t-statistic	4.51	-
p-value (two-tailed)	7.11×10^{-6}	-
Number of Months	1,190	-

Donec ultrices consectetur dolor sed sodales. Cras efficitur turpis vitae lacus mollis cursus. Pellentesque iaculis nulla ac vestibulum fermentum. Pellentesque et velit non turpis suscipit congue. Aliquam vitae purus a nibh cursus sagittis. Nullam egestas viverra purus vitae varius.

Sed maximus tempor vestibulum. Nunc tempor nulla justo, a posuere lorem tincidunt eget. Praesent dictum sed eros in pharetra. Vivamus nec dictum risus. Integer ac massa at magna rutrum pulvinar eget nec nunc. Curabitur tempor at lorem vitae venenatis. Suspendisse auctor posuere iaculis. Suspendisse egestas lobortis diam. Integer mattis.

2.2 Aggregate Commodity-Specific Factors

2.3 Individual Commodity Fundamentals

2.4 Macroeconomic Predictors

2.5 ML Methods in Return Forecasting

2.6 Volatility Regimes and Predictability

As you can see in Table 1.

3 Methodology

3.1 Baseline Framework (Replication)

We begin by gathering all T observations of n variables into a data matrix $\mathbf{X} \in \mathbb{R}^{T \times n}$, where each row is a *month* and each column is a macroeconomic variable. Each column is demeaned and standardized to unit variance using the StandardScaler:

$$X_{\text{scaled},ij} = \frac{X_{\text{raw},ij} - \mu_j}{\sigma_j}. \tag{1}$$

Let $\mathbf{\Sigma} = V[\mathbf{X}] \in \mathbb{R}^{n \times n}$ denote the variance-covariance matrix of $\mathbf{X}$, which is calculated as:

$$\mathbf{\Sigma} = \frac{1}{T-1}\mathbf{X}'\mathbf{X}, \tag{2}$$

and let $\mathbf{\Sigma} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}'$ be its spectral decomposition, where $\mathbf{U} \in \mathbb{R}^{n \times n}$ is an orthogonal matrix of order m, whose columns are the eigenvectors $\mathbf{u}_1, \mathbf{u}_2, ..., \mathbf{u}_n$ of $\mathbf{\Sigma}$, and $\mathbf{\Lambda} \in \mathbb{R}^{n \times n}$ is a diagonal matrix of the same order, whose diagonal entries are the corresponding eigenvalues $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_n$, which, by convention, are sorted from the largest to the smallest. By multiplying the original scaled data matrix $\mathbf{X}$ with $\mathbf{U}$, we derive the principal components of $\mathbf{X}$ as the elements of the matrix $\mathbf{F} \in R^{T \times n}$, or formally:

$$\mathbf{F} = \mathbf{X}\mathbf{U}. \tag{3}$$

To assess how much of the total variance in the original n variables is captured by the k-th principal component, or alternatively by the first K principal components, we can compute the variance-covariance matrix of $\mathbf{F}$ as:

$$\begin{aligned} \mathrm{V}[\mathbf{F}] &= \mathrm{V}[\mathbf{X}\mathbf{U}] \\ &= \mathbf{U}'\mathbf{\Sigma}\mathbf{U} \\ &= \mathbf{U}'\mathbf{U}\mathbf{\Lambda}\mathbf{U}'\mathbf{U} \\ &= \mathbf{I}_m\mathbf{\Lambda}\mathbf{I}_m \\ &= \mathbf{\Lambda}, \end{aligned} \tag{4}$$

where we exploited the orthogonality of $\mathbf{U}$. Recalling the fact that $\mathbf{\Lambda}$ is a diagonal matrix, $\mathrm{V}[\mathbf{F}] = \mathbf{\Lambda}$ implies that the principal components are mutually uncorrelated, and have a variance equal to the associated eigenvalue. Therefore, the proportion of variance in the original n variables explained by the k-th principal component is given by:

$$\frac{\lambda_k}{\sum_{i=1}^{n} \lambda_i}, \tag{5}$$

and the cumulative proportion of explained variance by the first K components by:

$$\frac{\sum_{k=1}^{K} \lambda_k}{\sum_{i=1}^{n} \lambda_i}. \tag{6}$$

Mirroring the approach used by Guidolin and Pedio, 2021, we forecast the returns of each commodity j, $R_{t+1,j}$, with the principal component analysis to summarize a large set of macroeconmic factors:

$$R_{t+1,j} = \underbrace{\alpha_j}_{\text{intercept}} + \underbrace{\sum_{i=1}^{N} \beta_{ij} PC_{t,ij}}_{\text{macroeconmic}} + \underbrace{D_{t,j} \left(\sum_{k=1}^{K} \gamma_{kj} C_{t,kj} \right)}_{\text{commodity-specific}} + \underbrace{\varepsilon_{t,j},}_{\text{error term}} \tag{7}$$

where α_j is the model's intercept,

3.2 Extension I: Penalized Regression

In the first alternate approach, we keep the principal component analysis and still extract ten principal components from the n inital variables, but instead of running stepwise AIC on those selected principal components, we run a LASSO regression as an alternative way to further reduce the number of parameters to be estimated. With this only being a narrow replacement compared to the original framework, every difference in results should be attributable soley to the selection method.

$$R_{t+1,j} = \alpha_j + \sum_{i=1}^{10} \beta_{ij}^{\text{LASSO}} PC_{t,ij} + \sum_{k=1}^{K} \gamma_{kj}^{\text{LASSO}} C_{t,kj} + \varepsilon_{t,j} \tag{8}$$

A broader extension to the approach by Guidolin and Pedio, 2021 is foregoing the principal component analysis entirely and applying LASSO directly to the n variables.

$$R_{t+1,j} = \alpha_j + \sum_{i=1}^{128} \beta_{ij}^{\text{LASSO}} X_{t,i} + \sum_{k=1}^{K} \gamma_{kj}^{\text{LASSO}} C_{t,kj} + \varepsilon_{t,j} \tag{9}$$

3.3 Extension II: Individual Fundamental Predictors

3.4 OOS Evaluation Criteria

3.5 Portfolio Back-Testing Framework

4 Data

4.1 Commodity Futures Returns

4.2 Macroeconomic Factors

4.3 Aggregate Commodity-Specific Factors

4.4 Individual Fundamental Predictors

4.5 Sample Splits and Descriptive Statistics

5 Replication & Baseline Results

5.1 In-Sample Estimation (1989-2003)

5.2 OOS Predictive Accuracy (2004-2012)

5.3 Portfolio Back-Testing (2004-2012)

5.4 Replication Assessment

6 Extension I: ML Variable Selection

6.1 Model Specification and Implementation

6.2 Variable Selection Patterns

6.3 OOS Forecasting Results - Original Sample (2004-2012)

6.4 OOS Forecasting Results - Extended Sample (2013-2024)

6.5 Portfolio Back-Testing

6.6 Robustness Checks

7 Extension II: Individual Commodity Fundamentals

7.1 Predictor Construction by Commodity Group

7.2 Individual vs. Aggregate Factors: Are They Substitutes?

7.3 OOS Forecasting Results (2004-2024)

7.4 Combined Model

7.5 Portfolio Back-Testing

7.6 Regime Interaction

8 Synthesis & Discussion

8.1 Model Horse Race

8.2 Statistical vs. Economic Loss Functions Revisited

8.3 Implications for Commodity Market Segmentation

8.4 Practical Implications

9 Conclusions

9.1 Summary of Findings

9.2 Limitations

9.3 Future Research Directions

References

Guidolin, Massimo and Manuela Pedio (Apr. 2021). “Forecasting Commodity Futures Returns with Stepwise Regressions: Do Commodity-Specific Factors Help?” In: *Annals of Operations Research* 299.1–2, pp. 1317–1356. ISSN: 0254-5330, 1572-9338. DOI: 10.1007/s10479-020-03515-w. URL: http://link.springer.com/10.1007/s10479-020-03515-w (visited on 04/17/2026).

Appendix